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Optimizing cooperation mechanisms for augmented reality (AR) services: Balancing product returns, pricing, and customer satisfaction

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ABSTRACT

Augmented Reality (AR) enhances user interaction with digital overlays, improving product marketing. This study investigates the interplay between AR services provided by influencers and product returns, focusing on two cooperation mechanisms: the agency mechanism (collaboration with suppliers) and the self-built store mechanism (independent operations with suppliers). Our research delves into the challenge of product returns and explores cooperation mechanisms between product suppliers and influencers utilizing AR services. The research results first show that higher customer satisfaction and providing AR services tend to increase product price and lead to higher return rates, while lower customer satisfaction and the absence of AR services result in lower product price but higher customer retention. Secondly, under the agency mechanism, when return loss costs are higher and customer satisfaction is lower, influencers provide more AR services. In the customer keep phenomenon, influencers should consider the wholesale price level, in addition to the unit management cost when deciding on high-quality AR services. Thirdly, under the agency mechanism, influencers choose to provide AR services only when customer satisfaction is very low but refrain in the self-built store mechanism. In addition, when customer satisfaction is higher, influencers can consider providing AR services and accepting returns. Furthermore, from the perspective of influencers, the agency mechanism is more advantageous when AR services are provided, while the self-built store mechanism is optimal without AR services under low satisfaction. High satisfaction consistently favors the agency mechanism.

1. Introduction

1.1. Background and motivation

With the rapid development of the e-commerce market, influencer marketing has become a pivotal force in driving industry growth (Liu et al., 2025). This marketing model enables direct interaction between product suppliers and consumers through influencers who leverage their significant online presence to boost sales and raise brand awareness (De Veirman et al., 2017; Singh et al., 2020; Weismueller et al., 2020). Collaborations with influencers provide suppliers not only with greater exposure and word-of-mouth promotion (Kim and Kim, 2021; Beichert et al., 2024) but also with a significant increase in sales revenue, positioning influencers as key players in the modern retail landscape (Nistor and Selove, 2024). However, there are various collaboration mechanisms available, such as influencer agencies and self-built online stores, each of which can affect the dynamics of the partnership (Liu and Wang, 2023; Li et al., 2023). This research contributes by examining how different influencer-supplier cooperation mechanisms can optimize marketing strategies, focusing on the emerging role of AR technology in enhancing influencer marketing and addressing challenges such as return rates and customer satisfaction.

Influencers can cooperate with companies through an agent mechanism to promote products, acting as intermediaries between the company and consumers (Doshi et al., 2023; Zheng et al., 2024). In this model, influencers promote products through sponsored posts, affiliate marketing, co-branded products, and giveaways on their social media channels (Zhang et al., 2023; Li et al., 2022; Li et al., 2023). Influencers typically earn a commission from the sales generated by their promotion efforts. However, customers may return products if they are dissatisfied, and in this agent-based mechanism, product suppliers are primarily responsible for shipping and handling returns (Li et al., 2022; Wang

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et al., 2022; Saha and Giri, 2023). For example, the cooperation between fashion brand Revolve and influencer Aimee Song has boosted Revolve's sales, with Revolve being responsible for returns and other related matters.[1] In addition, there is a new mechanism for influencers to collaborate by establishing their online stores, i.e., self-built store mechanism. Influential individuals on social media may choose to create online stores to sell products to their fans. This is known as the direct-to-consumer (DTC) mode, in which influencers are both the sellers and marketers of products. The process of influencers setting up their online stores typically includes product selection, supplier procurement, online store setup, marketing and promotion, and after-sales services. In general, in the cooperation mechanism for influencers' online stores, influencers are responsible for shipping and returns either by delivering products themselves or using a third-party logistics provider (Li et al., 2022; Wang et al., 2022; Saha and Giri, 2023). This mechanism requires more time and resources to establish and maintain an online store. For example, Kylie Jenner is one of the successful influencers who has used her online store to sell products. Initially, Kylie established her makeup brand through cooperation with a cosmetics manufacturer but later transitioned to the DTC model by opening her online store.[2] Therefore, by comparing the agency and self-built store mechanisms, this study aims to offer valuable insights into how influencers and suppliers can optimize their collaboration for more effective marketing and improved financial outcomes. Additionally, what role does the return cost play in the selection of these collaboration mechanisms?

Furthermore, Augmented Reality (AR) is revolutionizing influencer marketing by enhancing brand exposure and engagement through immersive and interactive experiences, such as virtual product try-ons and demonstrations (Kim and Park, 2024; Yu et al., 2024). For example, Nike's flagship stores utilize AR technology to elevate the shopping experience; customers can scan QR codes on products to access additional information, view virtual features, and try on shoes virtually. Influencers are increasingly leveraging AR to provide tailored and professional solutions for brands, thereby increasing their value and revenue potential (Hak, 2022; Ooi et al., 2023). These AR services not only enhance the influencer's appeal but also improve customer satisfaction by offering more accurate product experiences, which can lead to lower return rates (Ooi et al., 2023; Saha and Giri, 2023). For example, IKEA's IKEA Place AR app showcases how augmented reality can enrich customer experiences through influencer interactions. Therefore, our study will examine the role of AR in influencer-supplier collaborations, particularly focusing on its impact on customer return rates, product pricing strategies, and overall marketing effectiveness. By exploring how AR services influence consumer behavior, pricing decisions, and return policies, this research aims to provide insights into optimizing influencer marketing collaborations, ensuring greater consumer satisfaction and profits for both influencers and suppliers.

1.2. Research questions and methods

Building on the aforementioned background, the primary aim of this study is to explore the optimal cooperation mechanisms between influencers and product suppliers, with a particular focus on the role of Augmented Reality (AR) services. The study will seek to identify strategies that balance product pricing, return policies, and customer satisfaction. To achieve this, we will address the following key questions.

(i) How do customer returns affect product pricing and the level of Augmented Reality (AR) services under both the agency and self-built store mechanisms? What are the optimal pricing and AR service levels when customers either return products or retain them?
(ii) In the cooperation between influencers and product suppliers, under what conditions is it more advantageous for influencers to offer AR services to boost their income?
(iii) Which cooperation mechanism is more effective for influencers in marketing collaborations with product suppliers: the agency mechanism or the self-built store mechanism?

This research method integrates a multi-factor analysis framework, mathematical modeling, and game theory to study collaboration mechanisms, AR services, and product returns in influencer marketing. It provides rigorous tools for optimizing supply chain strategies, supporting pricing and service decisions while addressing gaps in multi-factor collaborative analysis in operations management.

Methods to address research gaps are as follows.

(i) Building a multi-factor integration analysis framework: This study unifies collaboration mechanisms (agent vs. self-built stores), AR services, and product returns into a comprehensive framework, addressing the limitations of prior research focused on isolated factors. It captures the interactions and combined effects of these variables, providing novel insights for optimizing influencer marketing strategies.
(ii) Application of mathematical modeling and quantitative analysis: A two-level supply chain model involving suppliers and influencers is constructed, with parameters such as customer satisfaction probability, return costs, and unit management costs. By quantifying variables like purchasing decisions, market demand, and returns, the study translates complex marketing phenomena into measurable relationships, enhancing rigor and practical applicability.
(iii) Application of game theory and strategic equilibrium analysis: Game theory analyzes strategy choices and equilibrium outcomes under different collaboration mechanisms. Sequential game models explore pricing, AR service provision, and service value decisions. This approach precisely characterizes profit structures, offering robust theoretical and quantitative insights into the impact of collaboration mechanisms on pricing, AR services, and profitability.

1.3. Research results and contribution statements

Through our exploration of the research objectives and questions, we have derived the following key findings. (i) We investigate product returns and customer retention in the market under the social influencer agency mechanism and self-built store mechanism. Our findings reveal that product returns are more likely to occur when customer satisfaction is high. Conversely, when customer satisfaction is low and the supplier incurs high return costs, customers tend to retain the products and avoid returning them. (ii) We analyze the differences in service provision by social influencers under the self-built store mechanism and agency mechanism in the context of product returns. In cases of returns, influencers under the agency mechanism offer higher AR services when the supplier's return costs are high and customer satisfaction is low. Conversely, under the self-built store mechanism, influencers provide higher AR services. In the customer retention scenario, when deciding whether to provide high-quality AR services, influencers need to consider both the unit management cost of products and the wholesale price level at the time of purchase. (iii) We examine the circumstances under which influencers, operating under the agency mechanism, would provide additional AR services for greater profits. Our analysis reveals that the service strategies adopted by influencers vary depending on the cooperation mechanism employed (agency mechanism or self-built store mechanism). When the cost of self-building a store is high, influencers may experience negative returns. (iv) In our analysis of influencers'

[1] https://www.jiemian.com/article/3171740.html.
[2] https://www.bbc.com/zhongwen/simp/business-44899547.

choice of cooperation mechanisms (self-built stores or agency mechanisms), we find that when providing AR services, influencers should opt for the agency mechanism. However, when customer satisfaction is low and influencers do not offer AR services, the self-built store mechanism is more suitable. When customer satisfaction is high, influencers should choose the agency mechanism.

We make some contributions as follows. Firstly, in the case of returns, influencers, under the agency mechanism, will provide higher AR services only when the product supplier's return loss cost is high and customer satisfaction is low. This also suggests that some influencers should consider the cost of returns when choosing a cooperation mechanism with product suppliers. In the case of the keep phenomenon, where consumers choose to retain the product, when deciding whether to provide high-quality AR services, influencers should not only consider the unit management cost of products but also focus on the level of wholesale price at the time of purchase. Secondly, under the agency mechanism, influencers can consider providing AR services only when customer satisfaction is very low. When customer satisfaction is high, they can consider providing no AR services and accepting customer returns. From the perspective of influencers under the self-built store mechanism, they can consider not providing AR services when customer satisfaction is very low. When customer satisfaction is high, they can consider providing AR services and accepting customer returns. Moreover, from the perspective of influencers under the self-built store mechanism, if the cost of building a store is high, they can choose to abandon the self-built store mechanism and choose the agency cooperation mechanism. This also suggests that influencers can choose different cooperation mechanisms based on whether they provide AR services. Thirdly, when providing AR services, influencers should choose the agency mechanism rather than the self-built store mechanism, which incurs higher costs. When not providing AR services, influencers should choose the self-built store mechanism when customer satisfaction is low; when customer satisfaction is high, influencers should choose the agency mechanism. This also suggests that influencers can consider whether they provide AR services when choosing different cooperation mechanisms.

The organization is as follows. Section 2 briefly reviews relevant literature. In section 3, we present the cooperation mechanism framework, model description, and sequence of events. In section 4, we carry on the model establishment, the solution, and the analysis. Section 5 presents the extended analysis by considering different commission ratios. The discussion section of Section 6 presents a comparison with previous research and management insights. Finally, we discuss the conclusion and future research in Section 7.

2. Literature review

2.1. Sales mechanisms for influencers

Influencer marketing and its impact on e-commerce have gained prominence. The research on influencer endorsements primarily focuses on using influencer brand promotion as a successful strategy in influencer marketing. Initially, some researchers viewed consumers as outsiders to influencer marketing, although the effectiveness of influencer endorsements is influenced by consumer psychology. However, recent studies have proposed mechanisms of influencer endorsements that highlight the interplay between the brand, the celebrity, and the consumer (Albert et al., 2017; Wang et al., 2025). Hani et al. (2018) and Juntiwasarakij (2018) highlight the global popularity of celebrity endorsements, while Hwang and Zhang (2018) and Jin (2018) focus on parasocial relationships and brand-related content on social platforms. Phua et al. (2018) and Zafar et al. (2019) examine congruence and authenticity in celebrity endorsements, using theories like Signaling Theory and cross-sectional regression. Geng et al. (2020) and Gilal et al. (2020) analyze economic impacts and brand passion, employing advanced econometric and psychological models. Hussain et al. (2020) and Park and Lin (2020) investigate celebrity trust and congruence effects on advertising and live-streaming outcomes. Teng et al. (2020) and Wongkitrungrueng and Assarut (2020) study live-streaming's influence on trust and customer engagement. More recently, Li et al. (2023) analyze the optimal sales mechanisms for live-streaming demonstration modes and the pricing strategies for an omnichannel manufacturer. Zhang et al. (2023) and Peng et al. (2023) explore live-streaming sales dynamics, while Chen et al. (2023) demonstrate retailers' advantages in adding live-streaming channels. Liu and Liu (2024) investigate AI-based influencer matching, and Che et al. (2025) and Liu et al. (2025) focus on brand-influencer fit and distribution strategies. Yang et al. (2025) examine anchor behavior's impact on sales, offering practical insights for e-commerce performance.

Research on the cooperation mechanism between product suppliers and influencers mainly focuses on the agency mechanism. Jiang and Cai (2021) create a pricing model for live e-commerce supply chains using online influencers as retailers. They examine the effects of impulsive buying behavior and satisfaction on consumers and analyze how such behavior affects manufacturers, retailers, and supply chains separately. Singh et al. (2021) propose an agent-based approach to examine the impact of influencers on idea generation. They support their model with the results of two empirical experiments that confirm the assumptions and logic used in the model. Doshi et al. (2023) introduce an agent-based model capable of simulating the dynamics of influencer advertising campaigns under various scenarios to determine the optimal influencer marketing strategy. Guo et al. (2024) investigate the best supply chain structure for a fixed-capacity hotel to add a live streaming channel when two hotels offer interchangeable rooms through the same online travel agency. Li et al. (2023) suggest a novel adaptive approach that combines agent-based modeling and genetic algorithms to tackle the challenging issue of identifying users' influence capability and optimizing influencer set selection in a dynamic environment. Their approach is based on an evolutionary framework and allows for distributed identification and optimization of influencer sets. Liu and Wang (2023) conduct research on the two business models, the Influencer-oriented Short-window model and the Market-oriented Long-window model, based on industrial practice. Their study aims to understand how the models benefit a firm and how decision-making affects profit sustainability and develop stochastic optimization models and derive managerial insights from their findings.

The distinction between this study and previous research lies in our consideration of Augmented Reality (AR) services provided by influencers in the context of different cooperation mechanisms (agency and self-built store) with product suppliers, alongside the presence of product returns in the market. We establish a model for product suppliers and influencers and analyze their pricing strategies, AR service levels, and profits through game theory under various cooperation mechanisms. Through quantitative analysis and numerical simulations, we identify the optimal cooperation mechanism for influencers.

2.2. Social influencer services

Currently, a large amount of research on social influencer services mainly focuses on surveys and empirical studies, which mainly explain social influencer services as a promotional tool and a new way of product promotion. Research on AR services demonstrates that perceived anthropomorphism in virtual influencers enhances user satisfaction (Kim and Park, 2024). Different factors influence purchase intentions based on AR experience: perceived usefulness is crucial for inexperienced users, while self-efficacy is key for experienced ones (Recalde et al., 2024). Additionally, emotions such as happiness, sadness, disgust, and surprise significantly impact user engagement with visually captivating content, highlighting the need to balance muscle movement intensity for optimal interaction between technology and human behavior (Yu et al., 2024). Studies also explore authenticity in virtual influencers, emphasizing diverse consumer and expert

perspectives (Koles et al., 2024), and identify convenience and informational value as critical for AR adoption, particularly in bridging the imagination gap (Schultz and Kumar 2024). Research on influencer services spans a variety of contexts, demonstrating their multifaceted impact on consumer behavior and business outcomes. Martinez-Lopez et al. (2020) conduct experimental studies to analyze the effects of brand control over shared messages, the commercial orientation of these messages, and the celebrity level and product congruence of influencers on social media. They define influencer marketing as compensating individuals for promoting products or services online. Shin and Lee (2021) study how apparel product type and the socioeconomic status of fashion influencers influence consumers' mimicry consumption processes, particularly from a curiosity perspective. Wu et al. (2022) explore the roles of influencers in service innovation, highlighting their resource integration behaviors with two firms. Hak (2022) examines the influence of influencers and live commerce services on brand image and purchase intentions, finding that these factors significantly impact consumer behavior for those who purchase products through influencer live commerce. Ooi et al. (2023) identify mobile convenience, interactivity, and influencer credibility as key factors shaping attitudes toward influencers and advertised products, which in turn affect actual purchase behavior. Bargoni et al. (2023) investigate how influencer endorsements affect funding intentions in crowdfunding campaigns, noting that platform trust and funder expertise play moderating roles. Guo et al. (2024) suggest that online travel agency live-streaming services may be more effective during economic downturns, as consumer travel intentions during recovery periods are influenced by platform reputation and live-stream presentation quality. Beichert et al. (2024) consider the entire influencer-marketing funnel, from follower engagement to actual revenue, while accounting for the costs of paid endorsements. Yi et al. (2025) analyze customer reviews of mobile stock trading apps to identify key satisfaction drivers, highlighting usability, information quality, and service quality, and introduce the Topic Rating Impact and Frequency Analysis (TRIFA) tool to visualize and improve service quality. Seo and Roh (2025) examine customer satisfaction in the O2O food delivery sector by distinguishing between online and offline service experiences, incorporating the technology acceptance model and trust transfer theory to explore how online satisfaction influences offline satisfaction and consumer loyalty. Building upon previous research on influencers' promotion services, this study examines different cooperation mechanisms between influencers and product suppliers (agency and self-built store). It specifically considers scenarios where influencers provide Augmented Reality (AR) services and where the market experiences product returns. The study establishes a model for product suppliers and influencers to determine the provision of AR services under different mechanisms. Utilizing game theory, it analyzes pricing strategies, AR service levels, and identifies the optimal cooperation mechanism for influencers, both with and without the provision of AR services.

2.3. Product returns

Recent research on product returns has primarily focused on several key areas. Firstly, product returns in dual-channel supply chains have garnered significant attention. Wang et al. (2022) analyze delivery distances, return periods, and pricing strategies using a two-stage optimization and Stackelberg game, comparing centralized and decentralized systems. Guo et al. (2022) investigate pricing and supply chain benefits in dual channels, incorporating offline returns and online reviews. Saha and Giri (2023) examine the role of return and warranty policies in influencing consumer choices between non-green and eco-friendly products. Huang et al. (2024) study the introduction of live-streaming channels, highlighting strategic impacts on competing retailers, including the potential for a prisoner's dilemma, mitigated by endogenous commissions. Liu et al. (2025) find that dual green innovations (process and recycling) synergistically enhance consumer demand and profitability, underscoring their importance for sustainable supply chains. Secondly, research on returns management strategies in supply chains has also been prominent. Li et al. (2022) use game theory to optimize returns management, comparing buyback and wholesale-price contracts. Feng et al. (2024) find that live social interactions in livestream e-commerce increase return rates, especially for experienced streamers and low-priced products. Nageswaran et al. (2024) reveal that return partnerships benefit online and store retailers with differentiated products but may backfire for similar offerings due to increased offline return rates for online purchases. Thirdly, the role of online and retail channels in influencing product returns has been a subject of interest. Letizia et al. (2018) highlight the complementary roles of online channels in reducing price marginalization and retail channels in minimizing returns. Xie et al. (2024) find chatbot humor enhances customer satisfaction via cognitive, emotional, and social pathways but is hindered by identity disclosure, offering guidance for chatbot design. Mrad et al. (2025) emphasize luxury product repairs in the circular economy, identifying barriers, drivers, and outcomes, and underscoring "slowing loop" strategies to extend luxury goods' lifecycles. Furthermore, several studies have focused on consumer behavior concerning return policies. Zhang and Choi (2021) analyze the impact of sales tax on multichannel strategies and social welfare, incorporating return policy implications. Cui et al. (2024) reveal that faster delivery promises to boost sales and profits but increase returns and lower retention, proposing a data-driven model to optimize delivery speed for customer lifetime value. Choi et al. (2024) explore the ship-then-shop model, showing how AI and predictive analytics optimize subscription fees and product prices while managing return impacts. Ghose et al. (2024) find that self-assurance nudges outperform pressure-based nudges, yielding higher long-term sales and fewer returns. Huang et al. (2024) study collection station capacity constraints, noting that stable or looser constraints mitigate the bullwhip effect, though strategic reductions can sometimes be advantageous.

This study builds upon existing research on product returns by considering various cooperation mechanisms (agency and self-built store) between influencers and product suppliers, as well as the impact of returns in the market. We establish models for influencers and product suppliers under different mechanisms, both with and without Augmented Reality (AR) services. Using game theory, we analyze pricing strategies, AR service levels, and identify the optimal cooperation mechanisms for influencers in scenarios involving product returns or keep.

3. Model framework and description

3.1. Problem description and assumption

As discussed in the background and literature review, social media influencers play a pivotal role in modern e-commerce, particularly in driving brand awareness and product sales through their large follower base and influence (Jiang and Cai, 2021; Guo et al., 2024). Studies indicate that collaborations between influencers and product suppliers are crucial in the contemporary retail landscape, especially regarding the impacts of agency and self-built store mechanisms on market performance (Liu et al., 2025; Saha and Giri, 2023). However, as consumer satisfaction fluctuates, return rates have become a key factor influencing the effectiveness of these collaboration mechanisms (Zhang et al., 2023). Against this backdrop, this paper aims to develop a two-tier supply chain model, consisting of a product supplier and a social media influencer, where the influencer is responsible for product promotion and sales on the social media platform. Specifically, the supplier provides the product, while the influencer takes on the responsibility of promoting and selling it, while also responding to market feedback, particularly regarding returns when consumers are dissatisfied (Peng et al., 2023). Based on the theoretical framework outlined above, this study will utilize a mathematical model to explore the collaboration mechanisms

between influencers and suppliers, focusing on how return costs affect pricing, product sales, and consumer satisfaction in different cooperation models. By constructing this model, we aim to analyze the impact of return costs under different collaboration mechanisms and uncover their operational efficiency. The next section will present a detailed description of the model used in this study, as well as define the key mathematical symbols involved, as shown in Table 1, to provide a theoretical foundation and methodological support for the subsequent analysis.

Cooperation Mechanisms. Currently, there are two main cooperation mechanisms between product suppliers and influencers. The first mechanism involves a social influencer agency model, where the social influencer is tasked with promoting the product. In this arrangement, the product supplier and the social influencer enter into a revenue-sharing agreement, where the influencer earns a percentage of the sales generated through their promotional efforts (Doshi et al., 2023). The product supplier retains responsibility for all logistics, including the delivery of the product to customers and the handling of any product returns. The full cost associated with product returns, including processing and restocking, is covered entirely by the product supplier. The second mechanism is the social influencer's self-built store model. In this arrangement, the product supplier sells the product in bulk to the social influencer at a wholesale price. The social influencer then takes full responsibility for operating their online store, including all aspects of product delivery to customers and handling returns. The costs associated with product returns, such as processing, shipping, and restocking, are entirely borne by the social influencer. Additionally, the social influencer is responsible for all expenses related to building, maintaining, and operating their self-built store, including any associated platform fees, marketing costs, and customer service operations.

Consumer. When customers make purchases through social influencers, the perceived value of the product by the customer, based on the social influencer's promotion, is denoted by v, v obeys [0,1] uniform distribution. And when customers receive the product, the probability of customer satisfaction for the product is ε, and then the consumer utility is v. Therefore, the probability of consumer dissatisfaction for the product can be expressed as: $1-\varepsilon$, where β captures the value discount from the dissatisfied product. Specially, $0 \le \varepsilon \le 1$ (Chen et al., 2020). We define the expected marginal valuation of the product as: $\chi = \varepsilon + (1-\varepsilon)\beta$ (Chen et al., 2020). The utility when the customer is satisfied with the product is: $U_s = v - p + \eta e$; the utility when the customer is dissatisfied with the product is: $U_{-s} = \beta v - p + \eta e$. Specially, e represents Augmented Reality (AR) services provided by influencers, p represents the price of the product, β represents the discount of product valuation when customers are not satisfied. And we considered whether the social influencer provides AR services. Therefore, we define $\eta = 0$ represents the scenario where the social influencer provides no additional AR services and $\eta = 1$ represents the scenario where they provide extra AR services.

Table 1
Definition of relevant parameters.

Notation	Definition
ε	The probability of customer satisfaction for the product.
z	The cost incurred by the customer during the return process of a product.
β	The value discount when customers are dissatisfied with the product.
c	The unit management cost of a product. The unit cost incurred by the social influencer in setting up their store, including inventory management, employee salaries, etc.
p	The retail price of the product.
w	The wholesale price of the product.
e	The AR service is provided by the social influencer.
n	The share of revenue that is shared between the product supplier and the social influencer in the revenue-sharing mechanism.
r	The cost of return loss, in the agency mechanism, is borne by the product supplier; in the self-built store mechanism, it is borne by the social influencer.
k	The cost coefficient of the AR service provided by influencers. To simplify the model, we assume that the cost coefficient of the AR service is $k = 1$.
F	The fixed cost of building a store for the social influencer.
π_s	The profit of the product supplier.
π_p	The profit of the social influencer.

Social influencer. The social influencer provides additional AR promotion services, such as virtual product try-ons, interactive demonstrations, and personalized shopping experiences. These AR services enhance customer engagement by offering a more immersive and accurate product experience, which can lead to higher customer satisfaction (Hak, 2022; Ooi et al., 2023). In addition, there are AR services such as KOL cooperation, where influencers collaborate with celebrities or other influencers on social media to promote products. This collaboration increases product exposure and sales opportunities through their fan bases or audience groups (Hak, 2022; Ooi et al., 2023). We define the basic service provided by influencers as e_0, the social influencer's AR service is e, the social influencer's all service is $e + e_0$. To simplify the model, we define $e_0 = 0$. The cost of the AR service provided by the social influencer is defined as: ke^2. Specially, k represents the cost coefficient for promotion services. Under the social influencer agency mechanism, the supplier's share of revenue is: $1-n$ (Doshi et al., 2023). The retail price of the product in the market is: p. In addition, under the self-built store mechanism, the return cost borne by the influencer to the consumer is rQ_{ri}, and the cost that influencers bear to build their stores is F. Specially, r is the cost of loss for each returned product, and Q_{ri} is the number of returned products.

Product supplier. The product supplier provides products, and under the social influencer agency mechanism, the supplier's profit share is n (Doshi et al., 2023). In the mechanism of influencers owning their stores, the wholesale price for the supplier is w. The retail price of the product in the market is: p. Under the social influencer agency mechanism, the return cost borne by the supplier to the consumer is rQ_{ri}. Specially, r is the cost of loss for each returned product, and Q_{ri} is the number of returned products.

Consumer purchase decision. Customer returns incur additional costs, such as restocking fees, handling charges, and potential loss of product value, all of which contribute to transaction expenses for both the seller and the consumer. In our study, z is the cost incurred by the customer during the return process of a product. When the customer is satisfied with the product, i.e. $v - p + \eta e > \eta e - z (v \ge v_1 = p - z)$, they will keep the product. In case of customer dissatisfaction with the product, i.e., $\beta v - p + \eta e \ge \eta e - z \left(v \ge v_2 = \frac{p-z}{\beta}\right)$, consumer purchase product. Therefore, when $v_1 \le v \le v_2$, the utility function for a customer who returns a product due to dissatisfaction after purchase is: $U_r = \varepsilon v + (1-\varepsilon)(p-z) - p + \eta e$; when $v_2 \le v \le 1$, the utility function of a customer who never returns a product is as follows (Chen et al., 2020): $U_{-r} = \chi v - p + \eta e$. Therefore, the utility function $E(U)$ for the customer is:

$$E(U) = \begin{cases} \eta e - z < 0, 0 \le v \le v_1. \\ \varepsilon v + (1-\varepsilon)(p-z) - p + \eta e, v_1 \le v \le v_2. \\ \chi v - p + \eta e, v_2 \le v \le 1. \end{cases} \quad (1)$$

For a clearer representation of the consumer valuation distribution, we summarize Fig. 1. Region A is comprised of consumers who place a very low value on the product, anticipating that the utility derived from a purchase will be non-positive; Region B encompasses consumers who will make a purchase but are likely to return the product if it does not meet their expectations; Region C consists of consumers who highly value the product, always choosing to buy and keep it.

When and only when the customer's utility $E(U) \ge 0 \left(i.e., v \ge \min\left\{p + \frac{(1-\varepsilon)z - e\eta}{\varepsilon}, \frac{p-\eta e}{\chi}\right\}\right)$, customers will only purchase a product when their utility is positive. When $p \ge \frac{-z(\chi-\beta)+\beta(e\eta-z)}{-(\chi-\beta)} \left(i.e., p \ge \frac{-z\beta+z(-1+\beta)\varepsilon+e\beta\eta}{(-1+\beta)\varepsilon}\right)$, $v_1 \le p + \frac{(1-\varepsilon)z-e\eta}{\varepsilon} \le \frac{p-\eta e}{\chi} \le v_2$, the market demand is

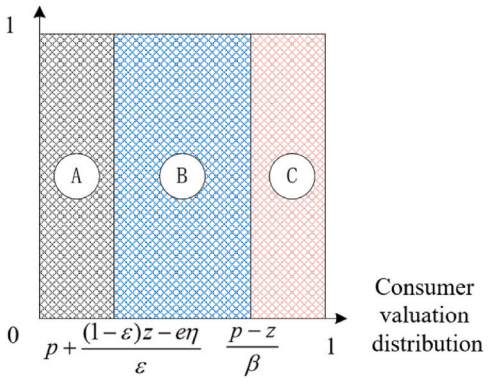


Fig. 1. Consumer valuation distribution.

$D_1 = 1 - p - \frac{(1-\varepsilon)z-e\eta}{\varepsilon}$. Because the estimated value $p+\frac{(1-\varepsilon)z-e\eta}{\varepsilon} \le v \le v_2$ of the returned product is $1-\varepsilon$, the quantity of products returned by the customer is $Q_{r1} = (1-\varepsilon)\left[\frac{p-z}{\beta} - p - \frac{(1-\varepsilon)z-e\eta}{\varepsilon}\right]$. When $0 \le p \le \frac{-z(\chi-\beta)+\beta(e\eta-z)}{-(\chi-\beta)}\left(i.e., 0 \le p \le \frac{-z\beta+z(-1+\beta)\varepsilon+e\beta\eta}{(-1+\beta)\varepsilon}\right)$, $v_1 \le v_2 \le \frac{p-\eta e}{\chi} \le p+\frac{(1-\varepsilon)z-e\eta}{\varepsilon}$, the market demand is $D_2 = 1-\frac{p-\eta e}{\chi}$. Because of $v_2 \le p+\frac{(1-\varepsilon)z-e\eta}{\varepsilon}$, the customer will not return the product, which means the quantity of returned products is $Q_{r2} = 0$. The market demand function can be summarized as follows:

$$D = \begin{cases} 1 - p - \dfrac{(1-\varepsilon)z-\eta e}{\varepsilon}, p \ge \dfrac{-z\beta+z(-1+\beta)\varepsilon+e\beta\eta}{(-1+\beta)\varepsilon}. \\ 1 - \dfrac{p-\eta e}{\chi}\left(i.e., 1 - \dfrac{p-\eta e}{\varepsilon+(1-\varepsilon)\beta}\right), otherwise. \end{cases} \quad (2)$$

The quantity of returned products is:

$$Q_{ri} = \begin{cases} (1-\varepsilon)\left[v_2 - p - \dfrac{(1-\varepsilon)z-e\eta}{\varepsilon}\right], p \ge \dfrac{-z\beta+z(-1+\beta)\varepsilon+e\beta\eta}{(-1+\beta)\varepsilon} \\ 0, otherwise \end{cases} \quad (3)$$

3.2. Sequence of event

This study uses a sequential game-theoretic approach, which is as follows (see Fig. 2).

- ***Step 1***: The product supplier decides which mechanism to work with the influencer (agency mechanism or self-built store mechanism).

In the social media influencer agent mechanism, the game sequence can be stated as follows.

- ***Step 2***: The product supplier determines the price p of the product in the social media influencer agency mechanism;
- ***Step 3***: In the social media influencer agency mechanism, the influencer decides whether to provide additional AR promotional services and if so, the influencer decides on the value of the AR services e provided.

In the mechanism of influencers building their stores, the game is played in the following order.

- ***Step 2***: The social influencer decides the product price p, and the social media influencer decides whether to provide additional AR promotional services and when the social media influencer decides to provide additional AR promotional services, the social media influencer determines the value of the AR services e.

We solve the game using backward induction and discuss it further in the Analysis section. Note that we derive technical restrictions on model parameters when ensuring second-order optimality conditions and considering parameter positivity constraints.

From a strategic perspective, this study investigates two potential cooperation mechanisms between product suppliers and social media influencers: the agency mechanism and the self-built store mechanism. The first step is to evaluate and identify the most advantageous mechanism for collaboration based on factors such as cost efficiency and profitability. Once the mechanism is chosen, the next strategic decision involves determining the optimal product pricing and whether

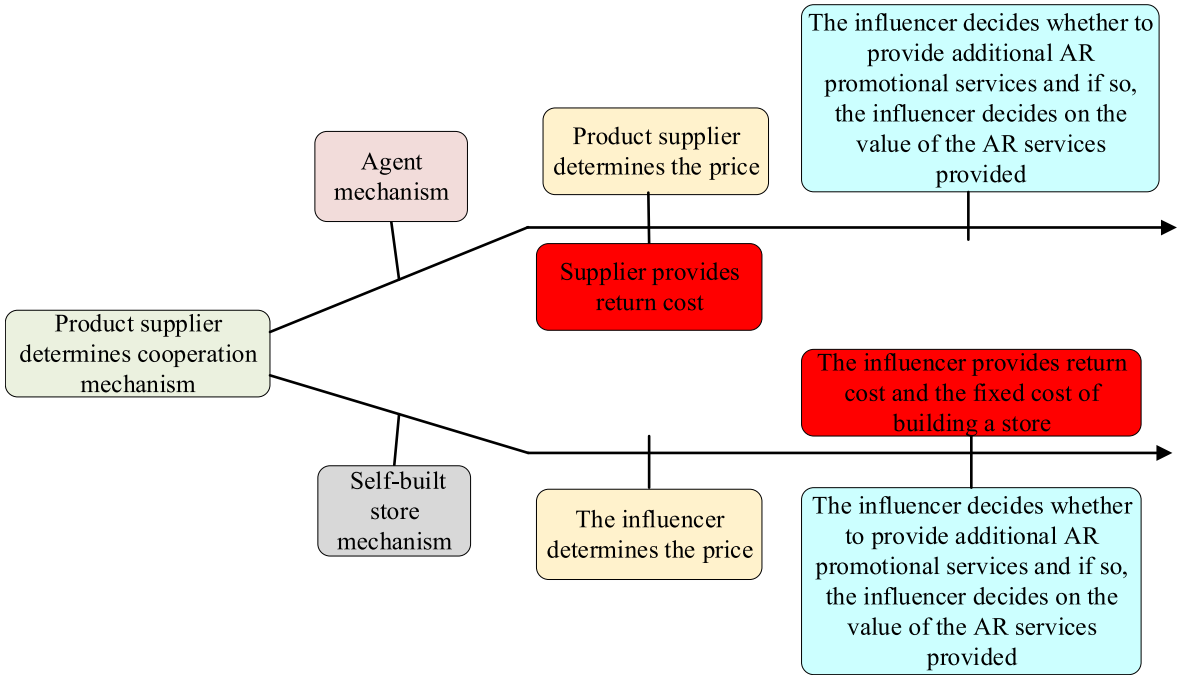


Fig. 2. Sequence of event.

influencers should offer additional AR promotional services. If AR services are included, the value of these services must also be strategically set to maximize consumer satisfaction and minimize return rates.

4. Analysis

4.1. The agency mechanism for social influencer

In the agency mechanism, based on whether consumers return the product ($i = 1$ or $i = 2$) and whether the social influencer provides AR services ($\eta = 1$ or $\eta = 0$), we have established a model to maximize profits for the product supplier and the social influencer. In addition, the "keep phenomenon" ($i = 2$) refers to consumers choosing to retain the product, while the "return phenomenon" ($i = 1$) refers to consumers opting to return the product. The profits of the supplier and influencer are as follows (Chen et al., 2020; Doshi et al., 2023; Guo et al., 2024).

According to the profit function expressions provided in Table 2, using game theory and the Kuhn-Tucker conditions method, the equilibrium results obtain under the social influencer agency mechanism are as follows:

Case I. Providing AR services in the return phenomenon (i.e., $i = 1, \eta = 1$).

$$p^{RI} = \frac{2n\beta\varepsilon(z(-1+\varepsilon)+\varepsilon) - r(-1+\varepsilon)((-1+n)\beta + 2(-1+\beta)\varepsilon^2)}{2n\beta(-1+n+2\varepsilon^2)}$$

$$e^{RI} = \frac{(-1+n)(-2n\beta\varepsilon(z(-1+\varepsilon)+\varepsilon)+r(-1+\varepsilon)((-1+n)\beta+2(-1+\beta)\varepsilon^2))}{4n\beta\varepsilon(-1+n+2\varepsilon^2)}$$

Case II. Providing AR services in the keep phenomenon (i.e., $i = 2, \eta = 1$).

$$p^{RII} = \frac{2z(\beta(-1+\varepsilon)-\varepsilon)\varepsilon}{(-1+n)\beta + 2(-1+\beta)\varepsilon^2}, e^{RII} = -\frac{(-1+n)z(\beta(-1+\varepsilon)-\varepsilon)}{(-1+n)\beta + 2(-1+\beta)\varepsilon^2}$$

Case III. No providing AR services in the return phenomenon (i.e., $i = 1, \eta = 0$)

$$p^{RIII} = \frac{1}{2}\left(1 + z - \frac{z}{\varepsilon} + \frac{r(-1+\beta+\varepsilon-\beta\varepsilon)}{n\beta}\right), e^{RIII} = 0$$

Case IV. No providing AR services in the keep phenomenon (i.e., $i = 2, \eta = 0$).

$$p^{RIV} = z + \frac{z\beta}{\varepsilon - \beta\varepsilon}, e^{RIV} = 0$$

Lemma 1. When $p \geq \frac{-z\beta+z(-1+\beta)\varepsilon+e\beta\eta}{(-1+\beta)\varepsilon}$, some customers in the market will return the product, while others will keep the product permanently.

Combined with Lemma 1, we also analyze the equilibrium prices for the four cases (**Case I-IV**) mentioned above and obtain the market phenomena shown in Fig. 3.

From Fig. 3, we can observe an interesting phenomenon. In **Region I**, when customers are highly satisfied, there may be a return of products in the market. In **Region II**, when customers are less satisfied and the product supplier's return loss cost is high, customers may keep the product and not return it. When customers are less satisfied and the product supplier's return loss cost is low, there may be a return phenomenon with the provision of AR services, and customers may hold the product without providing AR services. This is mainly because when customers are highly satisfied and AR services are provided, the price of the product is relatively high, so customers may choose to return the product. In Region III, when customers are less satisfied and no AR services are provided, the price of the product is relatively low, and there is a return inconvenience cost, so customers will choose to hold the product.

Table 2
The profits of the supplier and influencer.

	Return phenomenon ($i = 1$)	Keep phenomenon ($i = 2$)
Influencers provide AR services ($\eta = 1$)	$max_p\ \pi_s^{RI} = np^{RI}D_i^{RI} - rQ_{ri}$ $s.t.\ p \geq \frac{-z\beta+z(-1+\beta)\varepsilon+e\beta\eta}{(-1+\beta)\varepsilon}$ $max_e\ \pi_p^{RI} = (1-n)p^{RI}D_i^{RI} - ke^2$	$max_p\ \pi_s^{RII} = np^{RII}D_i^{RII}$ $s.t.\ p \geq \frac{-z\beta+z(-1+\beta)\varepsilon+e\beta\eta}{(-1+\beta)\varepsilon}$ $max_e\ \pi_p^{RII} = (1-n)p^{RII}D_i^{RII} - ke^2$
Influencers do not provide AR services ($\eta = 0$)	$max\ \pi_s^{RIII} = np^{RIII}D_i^{RIII} - rQ_{ri}$ $max_{p,e}\ \pi_p^{RIII} = (1-n)p^{RIII}D_i^{RIII}$ $s.t.\ p \geq \frac{-z\beta+z(-1+\beta)\varepsilon}{(-1+\beta)\varepsilon}$	$max\ \pi_s^{RIV} = np^{RIV}D_i^{RIV}$ $max_{p,e}\ \pi_p^{RIV} = (1-n)p^{RIV}D_i^{RIV}$ $s.t.\ p \geq \frac{-z\beta+z(-1+\beta)\varepsilon}{(-1+\beta)\varepsilon}$

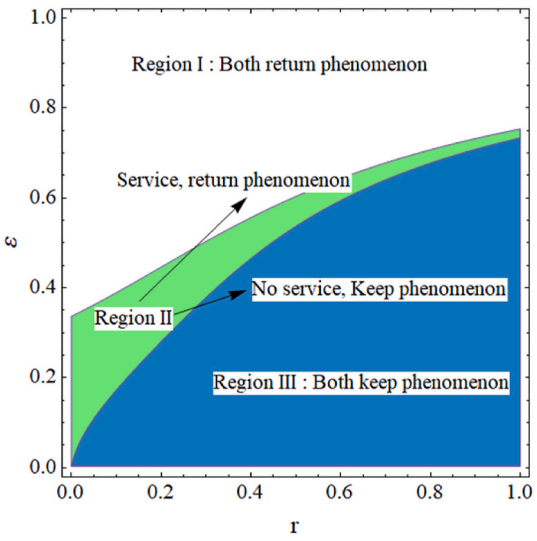


Fig. 3. Customer return behavior under the agency mechanism ($n = 0.5, \beta = 0.5, z = 0.1, \eta = 1$).

Proposition 1. The customer return cost z affects the price and level of AR promotion services of the product are as follows: (i) If $n > 1 - 2\varepsilon^2$, $\frac{\partial p^{RI}}{\partial z} < 0, \frac{\partial e^{RI}}{\partial z} < 0$; if $0 < n < 1 - 2\varepsilon^2, \frac{\partial p^{RI}}{\partial z} > 0, \frac{\partial e^{RI}}{\partial z} > 0$; (ii) $\frac{\partial p^{RII}}{\partial z} > 0, \frac{\partial e^{RII}}{\partial z} > 0$; (iii) $\frac{\partial p^{RIII}}{\partial z} < 0$; (iv) $\frac{\partial p^{RIV}}{\partial z} > 0$.

Proposition 1 shows the impact of customer return cost on product pricing and AR promotion services under the social influencer agency mechanism. Proposition 1(i) indicates that when the product supplier's profit share is relatively large, the price and AR promotion services of the product increase as the return cost increases when there is a returning phenomenon and services are provided; the opposite is true when the profit share is small. Proposition 1(ii) indicates that when there is a keep phenomenon (customers permanently keep the product) and services are provided, the price and AR promotion services of the product increase as the return cost increases. Proposition 1(iii) indicates that when there is a return phenomenon and no services are provided, the price of the product decreases as the return cost increases; the opposite is true when services are provided. Proposition 1(iv) indicates that when there is a keep phenomenon (customers permanently keep the product) and no services are provided, the price of the product increases as the return cost increases. Proposition 1 reveals that whether customers return products and whether influencers provide AR promotion services will both affect pricing and service levels under the agency mechanism.

Proposition 2. The influencer provides services that affect product pricing under the scenario of product returns as follows:

(i) If $n < 1 - 2\varepsilon^2$ and $0 < r < -\frac{n\beta}{-1+\varepsilon}$, when $0 < z < -\frac{r\varepsilon}{n\beta} - \frac{\varepsilon}{-1+\varepsilon}$, $p^{RIII} < p^{RI}$, when $z > -\frac{r\varepsilon}{n\beta} - \frac{\varepsilon}{-1+\varepsilon}$, $p^{RIII} > p^{RI}$;

(ii) If $n < 1 - 2\varepsilon^2$ and $r > -\frac{n\beta}{-1+\varepsilon}$, $p^{RIII} > p^{RI}$;

(iii) If $n > 1 - 2\varepsilon^2$ and $0 < r < -\frac{n\beta}{-1+\varepsilon}$, $p^{RIII} < p^{RI}$;

(iv) If $n > 1 - 2\varepsilon^2$ and $r > -\frac{n\beta}{-1+\varepsilon}$, when $0 < z < -\frac{r\varepsilon}{n\beta} - \frac{\varepsilon}{-1+\varepsilon}$, $p^{RIII} > p^{RI}$, when $z > -\frac{r\varepsilon}{n\beta} - \frac{\varepsilon}{-1+\varepsilon}$, $p^{RIII} < p^{RI}$.

Proposition 2 indicates that when the revenue sharing ratio is low, the supplier's return loss is low, and the consumer's return cost is low, the influencer's product pricing is lower when AR services are not provided. However, when the cost of returning goods to consumers is higher, the products are priced higher when the influencer does not offer AR services. When the revenue sharing ratio is low and the supplier's return loss is high, the influencer's product pricing is higher when no AR services are provided. When the revenue sharing ratio is high and the supplier's return loss is low, the influencer's product pricing is lower when AR services are not provided. When the revenue sharing ratio is higher, the supplier's return loss is higher, and the consumer's return cost is lower, the influencer's product pricing is higher when no AR service is provided. However, when the cost of returning goods to consumers is higher, the products are priced lower when the influencer does not provide AR services.

Proposition 3. The provision of AR promotional services affects the pricing of products in a keep scenario, i.e., $p^{RII} < p^{RIV}$.

Proposition 3 states that in the keep scenario under the social influencer agent mechanism, the price of a product when a social influencer provides AR promotional services is lower than when no such services are provided. This is mainly because the AR promotional services provided by the social influencer increase the product's added value, and customers who receive the product will not return it. Therefore, product suppliers will offer lower prices to attract more market consumers. This also suggests that if consumers enjoy the AR promotional services brought by influencers, and all market consumers do not return the product, it will lead to a lower purchase price.

4.2. The self-built store mechanism for influencers

In the self-built store mechanism, based on whether consumers return the product in the market ($i = 1$ or $i = 2$) and whether the social influencer provides AR services ($\eta = 1$ or $\eta = 0$), we have established a model to maximize profits for the product supplier and the social influencer. The profits of the supplier and influencer are as follows (Jiang and Cai, 2021; Guo et al., 2024; Wang et al., 2022; Chen et al., 2020):

According to the profit function expressions provided in Table 3, we applied game theory and the Kuhn-Tucker conditions method to obtain the equilibrium results under the self-built store mechanism, which are as follows:

Case I. Providing AR services in the return phenomenon (i.e., $i = 1, \eta = 1$).

$$p^{SI} = \frac{-\beta(c+r+w-r\varepsilon)+2\varepsilon(z\beta(-1+\varepsilon)+(1+c+w)\beta\varepsilon+r\varepsilon(-1+\beta+\varepsilon-\beta\varepsilon))}{\beta(-1+4\varepsilon^2)}$$

$$e^{SI} = \frac{z\beta(-1+\varepsilon)-(-1+c+w)\beta\varepsilon+r(1+\beta)(-1+\varepsilon)\varepsilon}{\beta(-1+4\varepsilon^2)}$$

Case II. Providing AR services in the keep phenomenon (i.e., $i = 2, \eta = 1$).

$$p^{SII} = \frac{\beta(c+w+z+\beta)+2z(-1+\beta)(\beta(-1+\varepsilon)-\varepsilon)\varepsilon}{2\left(\beta+(-1+\beta)^2\varepsilon^2\right)}, e^{SII} = \frac{2z\beta+(-1+\beta)(c+w-z+\beta)\varepsilon}{2\left(\beta+(-1+\beta)^2\varepsilon^2\right)}$$

Case III. No providing AR services in the return phenomenon (i.e., $i = 1, \eta = 0$).

$$p^{SIII} = \frac{z\beta(-1+\varepsilon)+(1+c+w)\beta\varepsilon+r\varepsilon(-1+\beta+\varepsilon-\beta\varepsilon)}{2\beta\varepsilon}, e^{SIII} = 0$$

Case IV. No providing AR services in the keep phenomenon (i.e., $i = 2, \eta = 0$).

$$p^{SIV} = z + \frac{z\beta}{\varepsilon-\beta\varepsilon}, e^{SIV} = 0$$

Table 3
The profits of the supplier and influencer.

	Return phenomenon ($i = 1$)	Keep phenomenon ($i = 2$)
Influencers provide AR services ($\eta = 1$)	$max_p\ \pi_s^{SI} = w^{SI}D_i^{SI}$ $s.t.\ p \geq \frac{-z\beta+z(-1+\beta)\varepsilon+e\beta\eta}{(-1+\beta)\varepsilon}$ $max_e\ \pi_p^{SI} = (p^{SI} - w^{SI} - c)D_i^{SI} - ke^2 - rQ_{ri} - F$	$max_p\ \pi_s^{SII} = w^{SII}D_i^{SII}$ $s.t.\ p \geq \frac{-z\beta+z(-1+\beta)\varepsilon+e\beta\eta}{(-1+\beta)\varepsilon}$ $max_e\ \pi_p^{SII} = (p^{SII} - w^{SII} - c)D_i^{SII} - ke^2 - F$
Influencers do not provide AR services ($\eta = 0$)	$max_p\ \pi_s^{SIII} = w^{SIII}D_i^{SIII}$ $s.t.\ p \geq \frac{-z\beta+z(-1+\beta)\varepsilon}{(-1+\beta)\varepsilon}$ $max_e\ \pi_p^{SIII} = (p^{SIII} - w^{SIII} - c)D_i^{SIII} - rQ_{ri} - F$	$max_p\ \pi_s^{SIV} = w^{SIV}D_i^{SIV}$ $s.t.\ p \geq \frac{-z\beta+z(-1+\beta)\varepsilon}{(-1+\beta)\varepsilon}$ $max_e\ \pi_p^{SIV} = (p^{SIV} - w^{SIV} - c)D_i^{SIV} - F$

Lemma 2. When $p \geq \frac{-z\beta+z(-1+\beta)\varepsilon+e\beta\eta}{(-1+\beta)\varepsilon}$, some customers in the market will return the product, while others will keep the product permanently.

Combined with Lemma 2, we analyze the equilibrium prices for the four cases (**Case I-IV**) mentioned above and obtain the market phenomena shown in Fig. 4.

We can observe a similarity between Figs. 3 and 4 in that the occurrence of returns and permanent ownership in the market depends on customer satisfaction. When customers are highly satisfied, the return of products may occur in the market; when customer satisfaction is low and the cost of product returns for the supplier is high, customers may choose to keep the product without returning it. Both Figs. 3 and 4 suggest that when customer satisfaction is high and AR services are provided, the product price is relatively high, and customers may choose to return the product; when customer satisfaction is low and AR services are not provided, the product price is relatively low, and with the cost of returns, customers may choose to keep the product.

Proposition 4. The customer return cost z affects the price and level of AR promotion services of the product are as follows: (i) If $\varepsilon \in \left(\frac{1}{2}, 1\right)$, $\frac{\partial p^{SI}}{\partial z} > 0, \frac{\partial e^{SI}}{\partial z} > 0$; if $\varepsilon \in \left(0, \frac{1}{2}\right), \frac{\partial p^{SI}}{\partial z} < 0, \frac{\partial e^{SI}}{\partial z} < 0$; (ii) $\frac{\partial p^{SII}}{\partial z} > 0$; $\frac{\partial e^{SII}}{\partial z} > 0$; (iii)

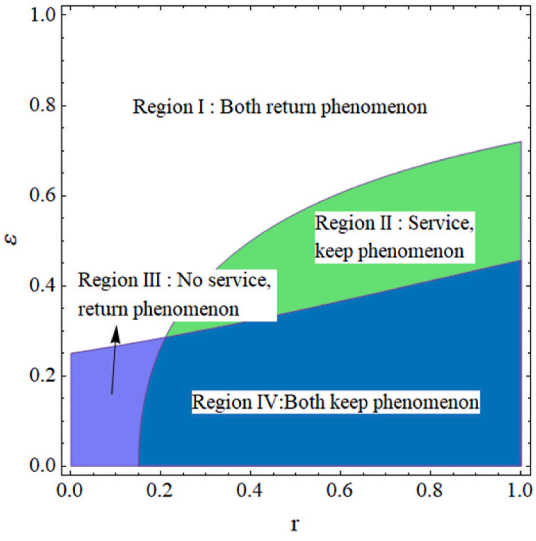


Fig. 4. Customer return behavior under self-built store mechanism.

$\frac{\partial p^{SIII}}{\partial z} < 0$; (iv) $\frac{\partial p^{SIV}}{\partial z} > 0$.

Proposition 4 states the impact of return costs on product pricing and AR promotional services under the social influencer self-built store mechanism. Proposition 4(i) states that when customer satisfaction is high, the price of the product and the AR promotional service increase as return costs increase in the presence of returns and provided services; the opposite is true otherwise. Proposition 4(ii) states that when there are keep phenomenon (customers hold the product permanently) and services are provided, the price and AR promotional services increase as return costs increase. Proposition 4(iii) states that when there are return incidents and no services are provided, the product price decreases as return costs increase; the opposite is true otherwise. Proposition 4(iv) states that when there are keep phenomenon incidents and no services are provided, the product price increases as return costs increase. Proposition 4 reveals that whether customers return products and whether influencers provide AR promotional services will both impact pricing and service levels under the social influencer self-built store mechanism.

Proposition 5. The provision of services affects product pricing in the context of product returns are as follows:

(i) If $0 < \varepsilon < \frac{1}{2}$ and $\vartheta(z,c,r) > 0$, $p^{SIII} < p^{SI}$;

(ii) If $0 < \varepsilon < \frac{1}{2}$ and $\vartheta(z, c, r) < 0$, when $0 < w < 1 - c + z + \frac{r(1+\beta)(-1+\varepsilon)}{\beta} - \frac{z}{\varepsilon}$, $p^{SIII} < p^{SI}$, when $w > 1 - c + z + \frac{r(1+\beta)(-1+\varepsilon)}{\beta} - \frac{z}{\varepsilon}$, $p^{SIII} > p^{SI}$;

(iii) If $\frac{1}{2} < \varepsilon < 1$ and $\vartheta(z, c, r) > 0$, when $0 < w < 1 - c + z + \frac{r(1+\beta)(-1+\varepsilon)}{\beta} - \frac{z}{\varepsilon}$, $p^{SIII} > p^{SI}$, when $w > 1 - c + z + \frac{r(1+\beta)(-1+\varepsilon)}{\beta} - \frac{z}{\varepsilon}$, $p^{SIII} < p^{SI}$;

(iv) If $\frac{1}{2} < \varepsilon < 1$ and $\vartheta(z,c,r) < 0$, $p^{SIII} < p^{SI}$.

Where $\vartheta(z,c,r) = \frac{z\beta(-1+\varepsilon)+\beta\varepsilon-c\beta\varepsilon+r(1+\beta)(-1+\varepsilon)\varepsilon}{2\beta\varepsilon(-1+4\varepsilon^2)}$.

Proposition 5 shows that factors such as product wholesale price, customer satisfaction, customer return cost and supplier return loss affect product pricing. Proposition 5 indicates that in the scenario of product returns under the social influencer's self-built store mechanism, when customer satisfaction is high or the supplier's loss cost is small and customer satisfaction is low, the product price is higher when the social influencer provides AR promotion services than when they do not. However, when the supplier's loss cost is high and customer satisfaction is high, the product price is lower when the social influencer provides AR promotion services than when they do not. This is mainly because the social influencer increases the added value of the product by providing AR promotion services, and as long as customer satisfaction is high, the product supplier will increase the product price.

Proposition 6. The effect of providing services on product pricing in a keep scenario are as follows: when $0 < z < \frac{(-1+\beta)(c+w+\beta)\varepsilon}{\beta(-2+\varepsilon)-\varepsilon}$, $p^{SII} > p^{SIV}$; when $z > \frac{(-1+\beta)(c+w+\beta)\varepsilon}{\beta(-2+\varepsilon)-\varepsilon}$, $p^{SII} < p^{SIV}$.

Proposition 6 suggests that in the keep scenario under the social influencer's self-built store mechanism when the cost of return is low for customers, the price of the product with AR promotion services provided by influencers is lower than that without the AR promotion services. This is mainly because influencers increase the added value of the product through AR promotion services, and customers are less likely to return the product, so product suppliers will lower the price to attract more market consumers. This also implies that if consumers enjoy the AR promotion services brought by influencers and no consumers return the product, then the purchase of the product only requires a lower price.

5. Comprehensive strategy analysis

5.1. Analysis of AR service with two cooperation mechanisms

Proposition 7. The differences in services provided by influencers under the self-built store mechanism and the agency mechanism in the return phenomenon are as follows: When $f(r,\varepsilon) > 0$, $e^{RI} > e^{SI}$; when $f(r, \varepsilon) < 0$, $e^{RI} < e^{SI}$.

Specially,

$$f(r,\varepsilon) = \frac{\begin{array}{c}\eta\left(r(-1+\varepsilon)\left(4\varepsilon^2-\eta^2\right)\left(2(-1+\beta)\varepsilon^2-\beta\eta^2\right)+n^2\left(4r(-1+\varepsilon)\varepsilon^2\eta^2+2z\beta(-1+\varepsilon)\varepsilon\left(4\varepsilon^2+\eta^2\right)+\beta\left(8\varepsilon^4-2(-1+2c+2w)\varepsilon^2\eta^2\right.\right.\right.\\ \left.\left.\left.-r\eta^4+r\varepsilon\eta^4\right)\right)+2n\left(r(-1+\varepsilon)\left(8\varepsilon^4+3(-1+\beta)\varepsilon^2\eta^2-\beta\eta^4\right)\right.\\ \left.-\beta\varepsilon\left((z(-1+\varepsilon)+\varepsilon)\eta^2+c\left(4\varepsilon^3-2\varepsilon\eta^2\right)+w\left(4\varepsilon^3-2\varepsilon\eta^2\right)\right)\right)\end{array}}{4n\beta\varepsilon(4\varepsilon^2-\eta^2)(2\varepsilon^2+(-1+n)\eta^2)}$$

In exploring the differences in services provided under the self-built store mechanism and the agency mechanism, we first analyze the differences in AR services provided by influencers in the return scenario (i.e., $n = 0.5, \beta = 0.5, z = 0.1, \eta = 1, c = 0.1, w = 0.2$). We find that the AR services provided by influencers under the agency mechanism were higher only when the product supplier's return loss cost was high and the customer satisfaction was low, while the AR services provided by influencers under the self-built store mechanism were higher in the opposite case (as shown in Fig. 5). This is mainly because, under the agency mechanism, the product supplier bears the cost of return losses if they are high, while under the self-built store mechanism, influencers bear the cost of return losses themselves, and the probability of returns will be higher when customer satisfaction is lower.

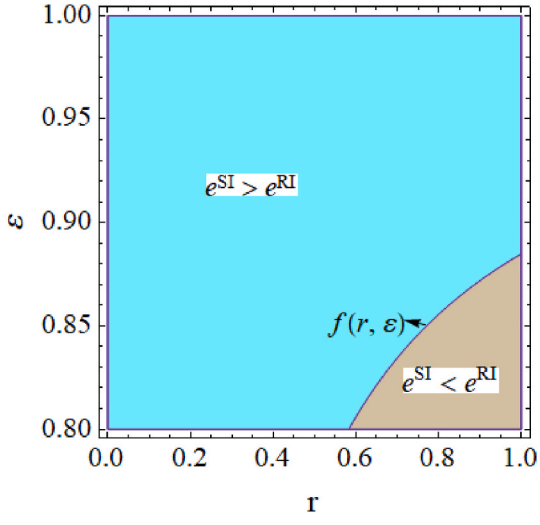


Fig. 5. Differences in services under the self-built store mechanism and the agency mechanism.

Therefore, from the perspective of influencers, the AR services provided by influencers under the agency mechanism are higher only when the product supplier's return loss cost is high and customer satisfaction is low. This also suggests that some influencers should fully consider the losses caused by returns when choosing a cooperation mechanism with product suppliers.

Next, we analyze the differences in the services provided by influencers in the self-built store mechanism and the agency mechanism under the condition of maintaining (i.e., keep phenomenon), as shown in Proposition 8.

Proposition 8. The differences in services provided by influencers under the self-built store mechanism and the agency mechanism in the keep phenomenon (i.e., keep phenomenon).

$$\begin{cases} c^* \le 0, e^{SII} < e^{RII}; \\ 0 < c^* < 1 \begin{cases} c \in (0, c^*), w \in (0, \min\{w^*, 1\}), e^{SII} > e^{RII} \\ c \in (0, c^*), w \in (\min\{w^*, 1\}, 1), e^{SII} < e^{RII} \\ c \in (c^*, 1), e^{SII} < e^{RII} \end{cases} \\ c^* \ge 1 \begin{cases} w \in (0, \min\{w^*, 1\}), e^{SII} > e^{RII} \\ w \in (\min\{w^*, 1\}, 1), e^{SII} < e^{RII} \end{cases} \end{cases}$$

Proposition 8 shows that when the unit management cost of a product is low and the wholesale price of the product under the self-built store mechanism is low, influencers under the self-built store mechanism provide higher AR services. Conversely, when the wholesale price of the product under the self-built store mechanism is high, influencers under the self-built store mechanism provide higher AR services. Interestingly, when the unit management cost of the product is high and the wholesale price of the product under the self-built store mechanism is low, influencers under the self-built store mechanism provide higher AR services. Conversely, when the wholesale price of the product under the self-built store mechanism is high, influencers under the self-built store mechanism provide higher AR services. This is mainly because when the wholesale price is low, influencers under the self-built store mechanism have lower procurement costs and are more willing to provide high-quality AR services. Therefore, when influencers decide whether to provide high-quality AR services, they should not only consider the unit management cost of the product, but also focus on the wholesale price when purchasing.

5.2. Optimal conditions for influencers to offer AR services

In exploring when influencers provide additional AR services for greater profits, we first analyze when influencers under the agency mechanism provide additional AR services for greater profits. We find that under the agency mechanism, influencers' provision of AR services yields higher profits when customer satisfaction is low and market customers do not return goods. When customer satisfaction is moderate, influencers under the agency mechanism do not provide AR services and yield higher profits when market customers return goods. However, when customer satisfaction is high, influencers under the agency mechanism do not provide AR services and yield higher profits when market customers return goods (as shown in Fig. 6). This is mainly because when customer satisfaction is lower, the market demand is lower, and only by providing AR services can the market demand increase and influencers' profits increase. When customer satisfaction is moderate or high, the cost of providing services outweighs the profits, and the influencers opt not to provide services. Additionally, when customer satisfaction is high, the possibility of returns can exist. Therefore, from the perspective of influencers, they should only consider providing AR services when customer satisfaction is very low under the agency mechanism. When customer satisfaction is moderate or high, they should consider not providing AR services and accepting customer returns.

Next, we analyze when influencers providing additional AR services would be more profitable under the self-built store mechanism. We find that when customer satisfaction is low, influencers under the self-built store mechanism do not provide AR services, and profits are higher when market customers do not return goods. However, when customer satisfaction is high, influencers under the self-built store mechanism provide AR services, and profits are higher when market customers return goods (as shown in Fig. 7). This is mainly because influencers under the self-built store mechanism need to bear the cost of both AR services and return losses. When customer satisfaction is low, the demand of market customers will be lower, and the profit of influencers is lower without providing AR services and when there is keep phenomenon. When customer satisfaction is high, the cost of providing services and returns will lead to a loss, and providing services to make profits can make up for the loss caused by returns. Therefore, from the perspective of influencers, influencers under the self-built store mechanism can consider not providing AR services when customer satisfaction is extremely low, and consider providing AR services and accepting customer returns when customer satisfaction is high.

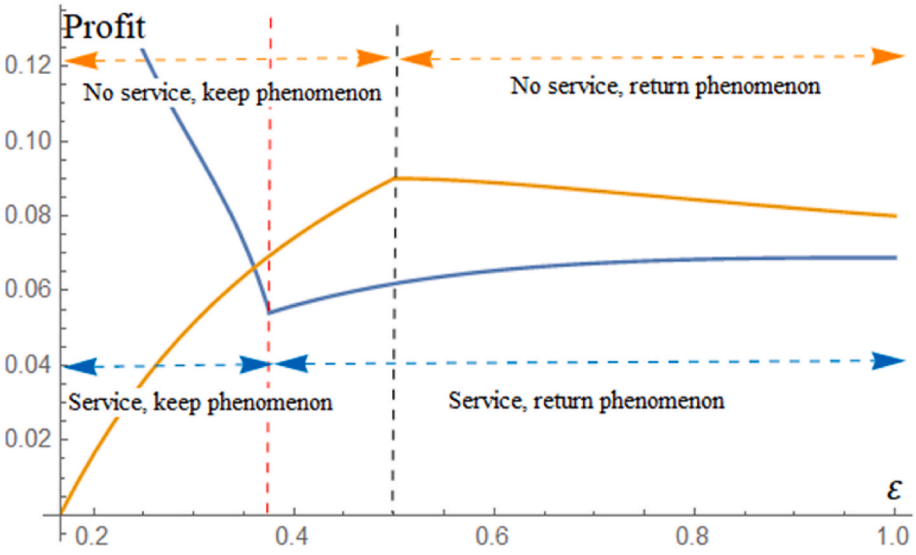


Fig. 6. Revenue of influencers providing additional AR services under the agency mechanism ($n = 0.5, \beta = 0.5, z = 0.1, c = 0.1, w = 0.2, r = 0.3$).

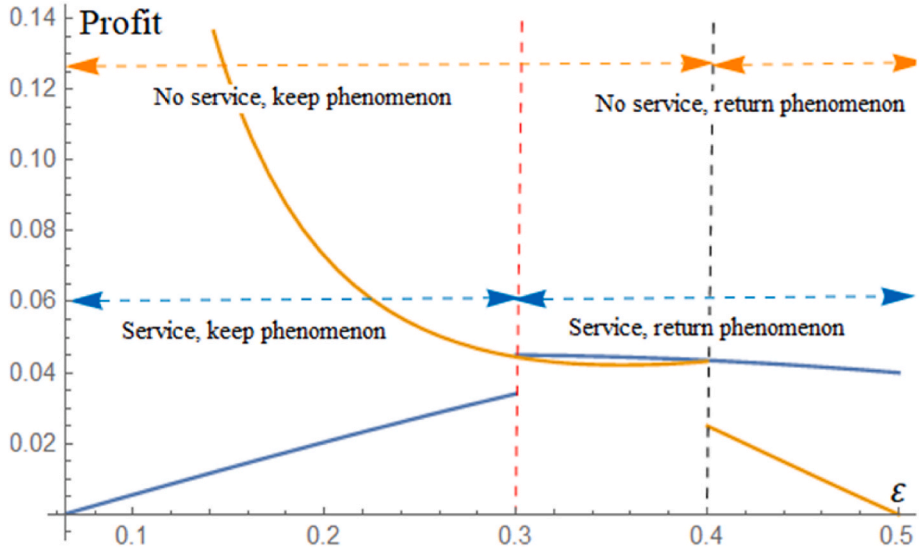


Fig. 7. The profits of influencers providing additional AR services under the self-built store mechanism (low fixed costs of the self-built store) ($\eta = 0, n = 0.5, \beta = 0.5, z = 0.1, r = 0.3, w = 0.2, c = 0.1, F = 0.000001$).

Finally, we also find from Figs. 6 and 7 that, from the perspective of influencers under the agency mechanism, influencers should consider providing AR services only when customer satisfaction is very low. When customer satisfaction is relatively high, they can consider not providing AR services and accepting customer returns. Under the self-built store mechanism, influencers should consider not providing AR services when customer satisfaction is very low, but should consider providing AR services and accepting customer returns when customer satisfaction is relatively high. We came to an interesting conclusion that when influencers adopt different cooperative mechanisms (agency mechanism and self-built store mechanism), the AR service strategies they provide are also different. This is mainly due to the difference in the cost of return loss under the two mechanisms. This also suggests that influencers can choose different cooperative mechanisms based on whether they provide AR services.

5.3. *Influencer cooperation mechanism choice: self-built stores or agency*

In our analysis of how influencers choose their cooperation mechanism (self-built stores or agency mechanisms), we first examined how influencers can maximize their profits by providing AR services under different cooperation mechanisms. We find that as customer satisfaction changes, influencers earn more profits by choosing the agency mechanism when providing AR services, whether the customer returns the product or not (as shown in Fig. 8). This is mainly because, in the self-built store mechanism, influencers bear more costs, such as the cost of providing AR services, the cost of return losses, and the fixed costs of building their stores. Therefore, from the perspective of influencers, when providing AR services, influencers should choose the agency mechanism instead of the self-built store mechanism, which incurs higher costs.

Next, we analyze how influencers can choose between self-operated stores or agency cooperation when they do not provide AR services. We find that when customer satisfaction is low, influencers should choose the self-operated store mechanism, while when customer satisfaction is high, they should choose the agency mechanism. Moreover, when customer satisfaction is high, the self-operated store mechanism yields negative returns (as shown in Fig. 9). This is mainly because when

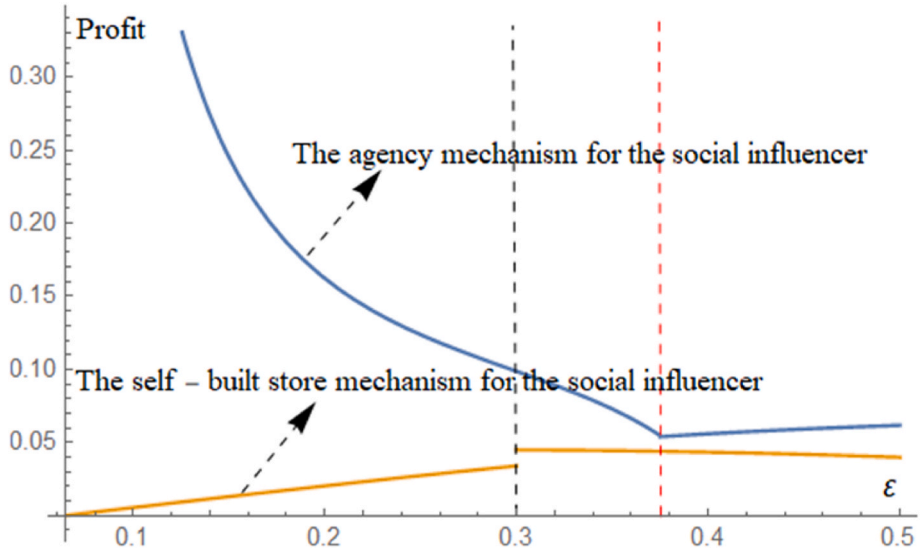


Fig. 8. Profits of different cooperation mechanisms for influencers providing AR services ($n = 0.5, \beta = 0.5, z = 0.1, \eta = 1, r = 0.3$).

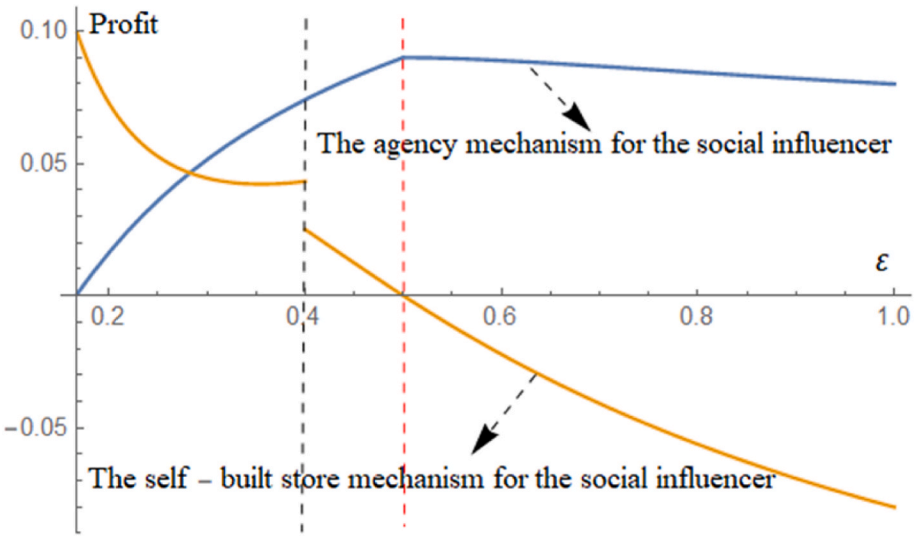


Fig. 9. Profits of influencers with different cooperation mechanisms when not providing AR services ($n = 0.5, \beta = 0.5, z = 0.1, \eta = 0, r = 0.3$).

influencers do not provide AR services, the self-operated store mechanism requires them to bear only the costs of return losses and fixed costs of self-operated stores. Therefore, from the perspective of influencers, when not providing AR services, they should choose the self-operated store mechanism when customer satisfaction is low, and the agency mechanism when customer satisfaction is high.

Furthermore, we find that, from the perspective of influencers, when providing AR services, influencers should choose the agency mechanism over the self-built store mechanism because influencers bear less cost compared to the self-built store mechanism, such as AR service costs, return loss costs, and fixed costs of self-built stores. On the other hand, when not providing AR services, influencers should choose the self-built store mechanism when customer satisfaction is low, and the agency mechanism when customer satisfaction is high. This suggests that influencers should consider whether they provide AR services when choosing different cooperation mechanisms.

6. Discussion

6.1. Comparison with existing studies

This study integrates AR services, product returns, and game theory within influencer marketing models, offering insights into pricing, AR services, and return management to optimize influencer-supplier collaboration. While previous research has focused predominantly on the agent model (Che et al., 2025; Liu et al., 2025), which positions influencers as intermediaries between brands and consumers, our study shows that influencers with their own sales platforms can leverage more control over product offerings, including AR service levels. This enables greater flexibility in pricing strategies and product positioning. For example, an influencer in the beauty industry who runs an online store might use AR to offer virtual try-ons for makeup products. By doing so, the influencer can increase consumer confidence in their purchase decisions, reducing the likelihood of product returns and boosting sales conversions. In this case, the integration of AR creates a more personalized shopping experience that ultimately leads to lower return rates and higher profitability.

Furthermore, the study introduces the role of AR in enhancing influencer marketing strategies. While previous research has explored AR's role in consumer decision-making (Kim and Park, 2024; Recalde et al., 2024), little has been done to connect this technology with influencer-marketing models. Our findings suggest that AR not only improves consumer engagement but also has strategic implications for pricing and product returns. By enabling consumers to virtually interact with products, influencers can increase trust and satisfaction, leading to fewer returns. For instance, in the fashion industry, an influencer might use AR technology to show how clothing fits various body types. This results in a more informed purchase decision, reducing the likelihood of returns due to size mismatches.

Another crucial contribution is the consideration of product return management within influencer marketing. Existing research on product returns has mainly focused on supply chains and e-commerce (Liu et al., 2025; Nageswaran et al., 2024), with limited exploration of how influencer models affect return rates. Our study demonstrates that integrating AR into the sales process can significantly reduce return rates by enhancing customer satisfaction and confidence in their purchases. For example, an influencer selling tech gadgets might provide AR-enabled demonstrations to show the product's functionality, helping customers make more informed decisions, thus reducing the chance of returns due to unmet expectations.

The application of game theory further enriches this study by analyzing how influencers and suppliers make strategic decisions regarding pricing, AR services, and returns. Previous game-theoretic models have focused on pricing and marketing decisions but have not fully considered how these decisions intersect with AR service levels or the management of returns (Jiang and Cai, 2021; Guo et al., 2024). Our study fills this gap by illustrating how these strategic elements interact within the influencer-supplier relationship. For instance, when influencers incorporate AR services, they not only offer a differentiated product experience but also influence the pricing strategy, making it possible to justify higher prices for products with enhanced virtual engagement features, while reducing the need for promotional discounts or return policies.

6.2. Management insights

Some management insights are as follows.

(1) **For product suppliers and influencers.** When customer satisfaction is high and AR services are provided, the product's pricing is relatively high, and there is a chance that customers will choose to return the product. When customer satisfaction is low and no AR services are provided, the product's pricing is relatively low, and customers have the inconvenience cost of returning the product, so they will choose to keep it.
(2) **For influencers.**

(i) AR service level. From the perspective of a social influencer in the context of product returns, the level of AR service provided by influencers in the agency mechanism is higher only when the product supplier's return loss cost is high and customer satisfaction is low. This also suggests that influencers should fully consider the losses caused by returns when choosing a cooperation mechanism with product suppliers. In the context of the keep phenomenon, when influencers decide whether to provide high-quality AR services, they need to focus not only on the unit management cost of the product but also on the wholesale price when purchasing.

(ii) When is it more profitable for influencers to provide additional AR services? From the perspective of influencers under the agency mechanism, influencers can consider providing AR services only when customer satisfaction is very low. When customer satisfaction is high, they can consider providing no AR services and accepting customer returns. From the perspective of influencers under the self-built store mechanism, influencers can consider not providing AR services when customer satisfaction is very low. When customer satisfaction is high, they can consider providing AR services and accepting customer returns. Furthermore, from the perspective of influencers, if the cost of building a store is high under the self-built store mechanism, they can abandon the self-built store mechanism and choose the agency mechanism. This also suggests that influencers can choose different cooperation mechanisms based on whether to provide AR services.

(iii) Influencers may choose between self-built stores and agency cooperation mechanisms. From the perspective of influencers providing AR services, when providing AR services, influencers should choose the agency mechanism rather than the self-built store mechanism because the agency mechanism incurs fewer costs. From the perspective of influencers not providing AR services, when customer satisfaction is low, influencers should choose the self-built store mechanism, while when customer satisfaction is high, influencers should choose the agency mechanism. This also suggests that influencers can consider their AR services when choosing different cooperation mechanisms.

7. Conclusion and future study

As observed in social influencer marketing, product suppliers (manufacturers) are increasingly establishing social influencer to attract consumers. In considering the different cooperation mechanisms between influencers and product suppliers, as well as consumer returns and social influencer AR services, we have established social influencer operation models under different cooperation mechanisms (agency mechanism and self-built store mechanism), filling the gap in the theoretical analysis of social influencer operation. The main conclusions of this study are summarized as follows.

Return phenomenon and keep phenomenon. When customers are highly satisfied, there is a phenomenon of product returns in the market. When customer satisfaction is low and the product supplier's return cost is high, there is a phenomenon of customers holding onto the product and not returning it. When customer satisfaction is low and the product supplier's return cost is low, there is a phenomenon of product returns with the provision of AR services and a phenomenon of customers holding onto the product without the provision of AR services.

The difference in services provided under the self-owned store mechanism and the agency mechanism. In the scenario of product returns, the influencers under the agency mechanism provide higher AR services only when the product supplier's return loss cost is high and customer satisfaction is low, whereas, under the self-built store mechanism, the influencers provide higher AR services. Conversely, in the scenario where customers do not return products, if the wholesale price of the product under the self-built store mechanism is low, the influencers under this mechanism provide higher AR services. Otherwise, if the wholesale price is high, the influencers under the self-built store mechanism provide higher AR services.

The profits of providing additional AR services by influencers. Under the agency mechanism, influencers provide AR services and earn higher profits when customer satisfaction is low and the market customers do not return products. When customer satisfaction is moderate, influencers under the agency mechanism do not provide AR services, and they earn higher profits when market customers return products. However, when customer satisfaction is high, influencers under the agency mechanism do not provide AR services, and they earn higher profits when market customers return products. Under the self-built store mechanism, when customer satisfaction is low, influencers do not provide AR services and earn higher profits when market customers do not return products. However, when customer satisfaction is high, influencers provide AR services under the self-built store mechanism and earn higher profits when market customers return products.

What cooperation mechanism do influencers adopt? We find that influencers have higher profits when providing AR services under the agency mechanism, regardless of whether customers return products or not, as customer satisfaction changes. When influencers do not provide AR services, we find that influencers have higher profits under the self-built store mechanism when customer satisfaction is low. When customer satisfaction is high, influencers have higher profits under the agency mechanism, and negative profits under the self-built store mechanism. Furthermore, at the level of influencers, when providing AR services, influencers should choose the agency mechanism over the self-built store mechanism, which bears more costs. When not providing AR services, influencers should choose the self-built store mechanism when customer satisfaction is low. When customer satisfaction is high, influencers should choose the agency mechanism.

This study offers valuable insights into the role of AR in influencer marketing but has several limitations. Firstly, it assumes deterministic service quality from influencers, yet the variability in service quality can significantly impact consumer behavior and the strategies employed by influencers and suppliers. Future research could address this by exploring stochastic models of AR service quality, focusing on its effects on pricing strategies and product returns. Secondly, the study mainly examines two cooperation models: agency and self-built store models. Expanding the analysis to include other models, such as affiliate marketing or subscription-based collaborations, could offer a more comprehensive understanding of influencer-supplier interactions. Empirical testing of these alternative mechanisms would enhance the literature on influencer marketing by shedding light on their influence on supplier strategies, pricing flexibility, and consumer trust. Additionally, the research assumes a single type of influencer, but influencers vary widely in reach, engagement, and market positioning. Future studies could explore how competition among different influencer types—such as micro, macro, and mega influencers—affects market outcomes like pricing, sales volume, and return policies. Investigating influencer competition could help suppliers select optimal influencers and devise more effective collaboration strategies.

CRediT authorship contribution statement

Zhitang Li: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis. **Peng He:** Writing – review & editing, Supervision. **Henry Xu:** Writing – review & editing, Methodology, Investigation.

Ethics approval and consent to participate

Not applicable (The manuscript does not report on or involve the use of any animal or human data or tissue. It used open access data which is not involving human participants, human data, or human tissue).

Consent for publication

Not applicable (Our manuscript does not contain data from any individual person).

Declaration of competing interest

The authors declare that they have no known competing interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jretconser.2025.104263.

Data availability

No data was used for the research described in the article.

References

Albert, N., Ambroise, L., Valette-Florence, P., 2017. Consumer, brand, celebrity: which congruency produces effective celebrity endorsements? J. Bus. Res. 81, 96–106.
Bargoni, A., Giachino, C., Battisti, E., Iaia, L., 2023. The effects of influencer endorsement services on crowdfunding campaigns. J. Serv. Market. 37, 40–52.
Beichert, M., Bayerl, A., Goldenberg, J., Lanz, A., 2024. Revenue generation through influencer marketing. J. Market. 88 (4). https://doi.org/10.1177/00222429231217471.
Che, S., Jin, X., Sheng, G., Lin, Z., 2025. Seeking effective fit: the impact of brand-influencer fit types on consumer brand attitude. J. Retailing Consum. Serv. 84, 104188.
Chen, Z.W., Fan, Z.P., Zhao, X., 2020. Offering return-freight insurance or not: strategic analysis of an e-seller's decisions. Omega-Int. J. Manag. Sci. 103.
Chen, Q., Yan, X., Zhao, Y., Bian, Y., 2023. Live streaming channel strategy of an online retailer in a supply chain. Electron. Commer. Res. Appl. 62, 101321.
Choi, W.J., Liu, Q., Shin, J., 2024. Predictive analytics and ship-then-shop subscription. Manag. Sci. 70 (2), 1012–1028.
Cui, R., Lu, Z., Sun, T., Golden, J.M., 2024. Sooner or later? Promising delivery speed in online retail. Manuf. Serv. Oper. Manag. 26 (1), 233–251.
De Veirman, M., Cauberghe, V., Hudders, L., 2017. Marketing through Instagram influencers: the impact of number of followers and product divergence on brand attitude. Int. J. Advert. 36, 798–828.
Doshi, R., Ramesh, A., Rao, S., 2023. Modeling influencer marketing campaigns in social networks. Ieee Transact. Computat. Soc. Sys. 10, 322–334.
Feng, X., Rong, Y., Tian, X., Wang, M., Yao, Y., 2024. When persuasion is too persuasive: an empirical analysis of product returns in livestream e-commerce. Prod. Oper. Manag., 10591478231224949
Geng, R., Wang, S., Chen, X., Song, D., Yu, J., 2020. Content marketing in e-commerce platforms in the internet celebrity economy. Ind. Manag. Data Syst. 120, 464–485.
Ghose, A., Lee, H.A., Nam, K., Oh, W., 2024. The effects of pressure and Self-Assurance nudges on product purchases and returns in online retailing: evidence from a randomized field experiment. J. Mark. Res. 61 (3), 517–535.
Gilal, F.G., Paul, J., Gilal, N.G., Gilal, R.G., 2020. Celebrity endorsement and brand passion among air travelers: theory and evidence. Int. J. Hospit. Manag. 85.
Guo, Q., Cheng, S., Zhao, P., 2024. How to introduce a livestreaming channel: a case of the hotel at a fixed capacity. Int. Trans. Oper. Res. 31 (5), 3531–3552.
Guo, X. G., Y. Liu & Z. J. Xia Decision analysis and coordination of dual supply chain with retailer's offline return service and online reviews. Manag. Decis. Econ. https://doi.org/10.1002/mde.3684.
Hak, L.J., 2022. The influence of influencer characteristics and live commerce service on brand image and consumer purchase intention. J. CEO Manag. Stud. 25, 197–215.
Hani, S., Marwan, A., Andre, A., 2018. The effect of celebrity endorsement on consumer behavior: case of the Lebanese jewelry industry. Arab Econ. Business J. 13, 190–196.
Huang, L., Liu, B., Zhang, R., 2024. Channel strategies for competing retailers: whether and when to introduce live stream? Eur. J. Oper. Res. 312 (2), 413–426.
Huang, S., Lu, H., Lin, J., Ponte, B., 2024. On the dynamics of return collection in closed-loop supply chains. Int. J. Prod. Res. 62 (3), 909–932.
Hussain, S., Melewar, T.C., Priporas, C.V., Foroudi, P., Dennis, C., 2020. Examining the effects of celebrity trust on advertising credibility, brand credibility and corporate credibility. J. Bus. Res. 109, 472–488.
Hwang, K., Zhang, Q., 2018. Influence of parasocial relationship between digital celebrities and their followers on followers' purchase and electronic word-of-mouth intentions, and persuasion knowledge. Comput. Hum. Behav. 87, 155–173.
Jiang, Y., Cai, H., 2021. The impact of impulsive consumption on supply chain in the live-streaming economy. IEEE Access 9, 48923–48930.
Jin, S.V., 2018. "Celebrity 2.0 and beyond!" Effects of Facebook profile sources on social networking advertising. Comput. Hum. Behav. 79, 154–168.
Juntiwasarakij, S., 2018. Framing emerging behaviors influenced by internet celebrity. Kasetsart J. Soc. Sci. 39, 550–555.
Kim, D.Y., Kim, H.Y., 2021. Influencer advertising on social media: the multiple inference model on influencer-product congruence and sponsorship disclosure. J. Bus. Res. 130, 405–415.
Kim, H., Park, M., 2024. When digital celebrity talks to you: how human-like virtual influencers satisfy consumer's experience through social presence on social media endorsements. J. Retailing Consum. Serv. 76, 103581.
Koles, B., Audrezet, A., Moulard, J.G., Ameen, N., McKenna, B., 2024. The authentic virtual influencer: authenticity manifestations in the metaverse. J. Bus. Res. 170, 114325.
Letizia, P., Pourakbar, M., Harrison, T., 2018. The impact of consumer returns on the multichannel sales strategies of manufacturers. Prod. Oper. Manag. 27 (2), 323–349.
Li, D., Chen, J., Chen, B.T., Liao, Y., 2022. Manufacturer's contract choice and retailer's returns management strategy. Transport. Res. E Logist. Transport. Rev. 165.
Li, W.H., Hu, Y.X., Jiang, C.T., Wu, S.Q., Bai, Q., Lai, E.D., 2023. ABEM: an adaptive agent-based evolutionary approach for influence maximization in dynamic social networks. Appl. Soft Comput. 136.
Li, Y., Li, B., Wang, M., Liu, Y., 2023. Optimal sales strategies for an omni-channel manufacturer in livestreaming demonstration trends. Transport. Res. E Logist. Transport. Rev. 180, 103222.
Liu, B., Wang, W., 2023. Live commerce retailing with online influencers: two business models. Int. J. Prod. Econ. 255.
Liu, J., Liu, Y., 2024. Asymmetric impact of matching technology on influencer marketing: implications for platform revenue. Mark. Sci. https://doi.org/10.1287/mksc.2023.0211.
Liu, J., Yuan, R., Feng, S., Wang, G., 2025. Quality-differentiated distribution strategies between live streaming and online channels. J. Retailing Consum. Serv. 82, 104056.
Liu, S., Yu, J.J., Feng, T., 2025. The impact of green innovations on firm's sustainable operations: process innovation and recycling innovation. Omega 130, 103170.
Martinez-Lopez, F.J., Anaya-Sanchez, R., Giordano, M.F., Lopez-Lopez, D., 2020. Behind influencer marketing: key marketing decisions and their effects on followers' responses. J. Market. Manag. 36, 579–607.
Mrad, M., Semaan, R.W., Christodoulides, G., Prandelli, E., 2025. Give me a second life! Extending the life-span of luxury products through repair. J. Retailing Consum. Serv. 82, 104055.
Nageswaran, L., Hwang, E.H., Cho, S.H., 2024. Offline returns for online retailers via partnership. Manag. Sci. 71 (1), 279–297.
Nistor, C., Selove, M., 2024. Influencers: the power of comments. Mark. Sci. https://doi.org/10.1287/mksc.2022.0186.
Ooi, K.B., Lee, V.H., Hew, J.J., Leong, L.Y., Tan, G.W.H., Lim, A., 2023. Social media influencers: an effective marketing approach? J. Bus. Res. 160.
Park, H.J., Lin, L.M., 2020. The effects of match-ups on the consumer attitudes toward internet celebrities and their live streaming contents in the context of product endorsement. J. Retailing Consum. Serv. 52.
Peng, J., Zhang, J., Nie, T., 2023. Social influence and channel competition in the live-streaming market. Ann. Oper. Res. 1–29.
Phua, J., Lin, J.-S., Lim, D.J., 2018. Understanding consumer engagement with celebrity-endorsed E-Cigarette advertising on instagram. Comput. Hum. Behav. 84, 93–102.
Recalde, D., Jai, T.C., Jones, R.P., 2024. I can find the right product with AR! The mediation effects of shopper engagement on intent to purchase beauty products. J. Retailing Consum. Serv. 78, 103764.
Saha, M., Giri, R.N., 2023. Consumers' purchasing decisions of a dual-channel supply chain system under return and warranty policies. Int. J. Sys. Sci.-Operat. Logistics 10.
Schultz, C.D., Kumar, H., 2024. ARvolution: decoding consumer motivation and value dimensions in augmented reality. J. Retailing Consum. Serv. 78, 103701.
Seo, K., Roh, T., 2025. Online and offline delivery qualities: O2O satisfaction and loyalty through technology acceptance model. J. Retailing Consum. Serv. 82, 104079.
Shin, E., Lee, J.E., 2021. What makes consumers purchase apparel products through social shopping services that social media fashion influencers have worn? J. Bus. Res. 132, 416–428.
Singh, H., Cascini, G., McComb, C., 2021. Influencers in design teams: a computational framework to study their impact on idea generation. Ai Edam-Artif. Intelli. Eng. Des. Analy. Manufact. 35, 332–352.
Singh, J., Crisafulli, B., Quamina, L.T., Xue, M.T., 2020. 'To trust or not to trust': the impact of social media influencers on the reputation of corporate brands in crisis. J. Bus. Res. https://doi.org/10.1016/j.jbusres.2020.03.039.
Teng, W., Su, Y., Liao, T.-T., Wei, C.-L., 2020. An exploration of celebrity business ventures and their appeal to fans and non-fans. J. Retailing Consum. Serv. 54.
Wang, B., Han, Y., Kandampully, J., Lu, X., 2025. How language arousal affects purchase intentions in online retailing? The role of virtual versus human influencers, language typicality, and trust. J. Retailing Consum. Serv. 82, 104106.
Wang, Q., Wu, J., Zhao, N.G., 2022. Optimal operational policies of a dual-channel supply chain considering return service. Math. Comput. Simulat. 199, 414–437.
Weismueller, J., Harrigan, P., Wang, S., Soutar, G.N., 2020. Influencer endorsements: how advertising disclosure and source credibility affect consumer purchase intention on social media. Australas. Mark. J. https://doi.org/10.1016/j.ausmj.2020.03.002.
Wongkitrungrueng, A., Assarut, N., 2020. The role of live streaming in building consumer trust and engagement with social commerce sellers. J. Bus. Res. 117, 543–556.

Wu, Y., Nambisan, S., Xiao, J.H., Xie, K., 2022. Consumer resource integration and service innovation in social commerce: the role of social media influencers. J. Acad. Market. Sci. 50, 429–459.
Xie, Y., Liang, C., Zhou, P., Jiang, L., 2024. Exploring the influence mechanism of chatbot-expressed humor on service satisfaction in online customer service. J. Retailing Consum. Serv. 76, 103599.
Yang, Y., Zhao, J., Li, Y., 2025. More than enough is too much: curvilinear relationship between anchor body movements and sales in live streaming e-commerce. J. Retailing Consum. Serv. 84, 104211.
Yi, J., Oh, Y.K., Kim, J.M., 2025. Unveiling the drivers of satisfaction in mobile trading: contextual mining of retail investor experience through BERTopic and generative AI. J. Retailing Consum. Serv. 82, 104066.
Yu, J., Dickinger, A., So, K.K.F., Egger, R., 2024. Artificial intelligence-generated virtual influencer: examining the effects of emotional display on user engagement. J. Retailing Consum. Serv. 76, 103560.
Zafar, A.U., Qiu, J., Li, Y., Wang, J., Shahzad, M., 2019. The impact of social media celebrities' posts and contextual interactions on impulse buying in social commerce. Comput. Hum. Behav. https://doi.org/10.1016/j.chb.2019.106178.
Zhang, T., Choi, T.M., 2021. Optimal consumer sales tax policies for online-offline retail operations with consumer returns. Nav. Res. Logist. 68 (6), 701–720.
Zhang, W., Yu, L., Wang, Z., 2023. Live-streaming selling modes on a retail platform. Transport. Res. E Logist. Transport. Rev. 173, 103096.
Zheng, L., Huang, B., Qiu, H., Bai, H., 2024. The role of social media followers' agency in influencer marketing: a study based on the heuristic–systematic model of information processing. Int. J. Advert. 43 (3), 554–579.